

Power envelopes, exact defects, and cycle restrictions for the Juggler map

Short note. Not submitted.

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

We develop an exact finite-word calculus for the Juggler map. For every parity word w realized at n , the endpoint satisfies the power envelope

$$J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Hence $3^{\#O(w)} < 2^{|w|}$ forces $J^{|w|}(n) < n$ whenever $n \geq 2$. Local floor remainders lift through later exponents to an exact global defect. We prove its identity, vanishing classification, and composition law: concatenation is a two-term power-gap, and zero defect gives precisely the rigid monochrome towers. Exact inverse cells then yield necessary cycle restrictions, including superquadratic prefixes from a cycle minimum to later even states, and a small-cycle census: the Juggler map has no nontrivial cycle of length at most six. Length seven remains open.

As a secondary application, every even start and every odd start whose first image is even has a uniform one- or two-step descent certificate. The starts not covered by those two certificates are exactly the odd-to-odd class; many of those starts still descend after a longer word. No density result is stated or used here.

All finite-word statements are conditional on the realized itinerary.

1. Introduction

The Juggler sequence was introduced by Pickover [1]; see also OEIS A094683 [2]. Universal arrival at 1 remains open.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \quad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities. The paper records the exact finite-word comparison, the compositional slack, and the uniform short certificates.

The main contribution is the exact power-envelope and defect calculus, together with its inverse-cell and cycle consequences, chief among them a small-cycle census: no nontrivial cycle has length at most six. To the best of our knowledge, the global-defect identity and composition law (Theorems 2.4 and 2.6) and the cycle census (Theorem 3.6) are new for the exact Juggler map. Section 2 records the envelope and defect calculus. Section 3 proves the cycle restrictions and the census. Section 4 records the uniform short certificates. Section 5 states the remaining gap.

1.1 Verification convention

The arguments of Sections 2–4 are written below and may be read without Lean. Lean supplies an independent check of those arguments; the corresponding names are collected in Appendix A. Lemma 3.5 uses a table of 254 six-step evaluations for $2 \leq n < 256$; that table is a finite computation, not a termination proof. The four-block chain in Section 5 is Lean-certified.

1.2 Related work

The map is Pickover’s [1]. The one-step sequence is OEIS A094683 [2]; the stopping-time table is A007320 [8]. We know of no published exclusion of nontrivial cycles for J . Prasad–Prasad [9] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to the exact floor-power map. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [3,4]; those results do not apply here, because the branches are floor powers rather than affine maps (Crandall [5], Matthews–Watts [6]). Itinerary-class densities are the subject of a companion manuscript in preparation [7], which imports from this note only Corollary 2.3; the present arguments do not use it.

2. Finite words, envelope, and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w . Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters.

The identities in this section are formalized in Lean; names are in Appendix A. The proofs below are the ordinary integer arguments.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w , then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w . The empty word is immediate. For the induction step, the two current states realize the same next letter. On the even branch, monotonicity follows from monotonicity of the integer square root; on the odd branch it follows from monotonicity of $x \mapsto x^3$ followed by the integer square root. Apply the induction hypothesis to the prefix and then the appropriate branch inequality. $\square$

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. Write $k = |w|$ and $o = \#O(w)$. The empty word is the equality $n \leq n$. Suppose the bound holds at a realized prefix ending at x , and the next letter is realized.

If the next letter is even, then $J(x) = \lfloor \sqrt{x} \rfloor$, so $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{2^k} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x) = \lfloor x^{3/2} \rfloor$, so $J(x)^2 \leq x^3$. Therefore

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{3 \cdot 2^k} = (x^{2^k})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. $\square$

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n , and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. $\square$

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting word.

The floor slack is exact, and it is not an additive path sum. For a single branch,

$$x^e = J(x)^2 + \rho(x), \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write

$$\text{gap}(a, \rho, e) = (a + \rho)^e - a^e,$$

so $a^e + \text{gap}(a, \rho, e) = (a + \rho)^e$. If $e \geq 1$, then $\text{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \leq \text{gap}(a, \rho, e)$.

Along a realized word w , write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \text{gap}(x_{j+1}^2, \rho_j, 2^j), & \text{next letter even,} \\ \text{gap}(x_{j+1}^2, \rho_j, 2^j) + \text{gap}(x_j^{2^j}, D_j, 3), & \text{next letter odd.} \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j . An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length k with odd count o ends at x and satisfies $n^{3^o} = x^{2^k} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^k} + D = J(x)^{2^{k+1}} + D + \text{gap}(J(x)^2, \rho(x), 2^k).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^k} + D)^3 = x^{3 \cdot 2^k} + \text{gap}(x^{2^k}, D, 3).$$

The leading term is

$$x^{3 \cdot 2^k} = (J(x)^2 + \rho(x))^{2^k} = J(x)^{2^{k+1}} + \text{gap}(J(x)^2, \rho(x), 2^k),$$

which is the odd update. $\square$

For a one-letter illustration, take $n = 3$ and $w = O$. Then $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$. The local remainder is $\rho = 2$, and the global defect equals that remainder only because the word has length one; after a later letter the lift is a power-gap, not a sum of remainders.

Theorem 2.5 (vanishing). If w is realized at n , the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.
3. $J^{|w|}(n)^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a , or $w = O^k$ and $n = a^{2^k}$ for an odd a . A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives $(1) \Leftrightarrow (3)$. For $(1) \Leftrightarrow (2)$, use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is divisible by 2. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k , hence $n = a^{2^k}$ for an odd a . Conversely, these even and odd towers make every local remainder zero. $\square$

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{\#O(v)}) + \text{gap}(J^{|v|}(m)^{2^{|v|}}, \Delta_v(m), 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = (m^{2^{|u|}} + \Delta_u(n))^{3^{o_v}} = J^{|uv|}(n)^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|} \cdot 3^{o_v}} + \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$(J^{|v|}(m)^{2^{|v|}} + \Delta_v(m))^{2^{|u|}},$$

which expands as the second gap plus $J^{|uv|}(n)^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \leq \text{gap}(a, \rho, e)$ on the second summand. For a local remainder at index j , split w after j letters: the suffix defect is at least ρ_j , and raising through 2^j gives the stated bound. $\square$

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power, and later letters of either parity raise a suffix slack through $2^{|u|}$. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n , that is $J^{|w|}(n) = n$, then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly. A cycle burns its entire formal surplus in floor losses.

Proof. Theorem 2.4 with $m = n$. Nonnegativity of $\Delta_w(n)$ makes the difference a genuine one. $\square$

The identity does not supply a state-independent positive tax, and none exists: the single-step slack is provably squeezed by the scale of the state. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2$$

so the relative slack $1 + \eta = x^e / J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. A uniform per-step tax is therefore impossible. The recorded extreme is an *OOE* block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose word-relative slack

$$q_w(n) = \frac{n^{3^{\#O(w)}}}{J^{|w|}(n)^{2^{|w|}}} - 1$$

satisfies $0 < q_{OOE}(n) < 10^{-30}$; this is an exact-integer computation, not an estimate:

```
from math import isqrt

n = 180370579261640036336071806107777
def J(x):
    return isqrt(x) if x % 2 == 0 else isqrt(x**3)
x = J(J(J(n))) # OOE realized; x = 1941719144218166368455510841464890645
print(0 < (n**9 - x**8) * 10**30 < x**8) # True
```

The inequality $\Delta_w(n) > n^{3^{\#O(w)}} - n^{2^{|w|}}$ on a formally expanding word is exactly $J^{|w|}(n) < n$, so it does not forbid mixed expanding prefixes.

3. Inverse cells and cycles

The exact one-step fibers are

$$J(n) = q \iff q^2 \leq n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \leq n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd cells are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd cell indexed by m . Then $(a+1)^3 \leq b^3 < (m+1)^2$ and $m^2 \leq a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

and hence $3a^2 \leq 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \leq a^3$ gives $9a^4 \leq 4m^2 \leq 4a^3$, contradicting $4a^3 < 9a^4$. $\square$

A nonempty realized word w with $J^{|w|}(n) = n$ is a cycle word. The identities in this section are formalized in Lean; names are in Appendix A. The proofs below are the ordinary integer arguments.

Theorem 3.2 (cycle restrictions). Let w be a cycle word at $n \geq 2$.

- (i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting word cannot close a nontrivial cycle.

- (ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

- (iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix *OOE* is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n . If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n . Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. $\square$

Lemma 3.3 (coarse lower envelope). For $n \geq 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \leq n < (q+1)^2$. For $q \geq 1$ one has $(q+1)^2 \leq 4q^2$, and the second inequality is strict for $q \geq 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \leq 4J(n)^2$ and an odd step satisfies $n^3 \leq 4J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq C_v J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j . In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. $\square$

Lemma 3.4 (next-square thresholds). (i) If $q \geq 5$ realizes OO , then $J^2(q) \geq (q+1)^2$. (ii) If $q \geq 3$ realizes OOO , then $J^3(q) \geq (q+1)^2$. (iii) If a realized word v satisfies $J^{|v|}(q) \geq (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \geq (q+1)^2$. (iv) If vE is a cycle word at n , then $J^{|v|}(n) < (n+1)^2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \geq (q+1)^2$ follows from $m^3 \geq (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \geq (q+1)^2$. If $q = 5$, then $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \geq 6^4 = 1296$. Since q realizes OO , it is odd, so the only remaining case is $q \geq 7$. Then $m \geq q^{3/2} - 1$, so it is enough that $(q^{3/2} - 1)^3 \geq (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to $q^{9/2} - 3q^3 \geq (q+1)^4$, or equivalently $\sqrt{q} - 3/q \geq (1 + 1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the orbit $3 \rightarrow 5 \rightarrow 11 \rightarrow 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO , then it realizes OO , so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$. $\square$

Lemma 3.5 (two length-six exclusions). Neither $OOOEOE$ nor $OOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ because $(1 + 1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n+1) \leq 257n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $2 \leq n < 256$, neither word returns to its start. This is a table of 254 evaluations of a six-step integer map and a return test. The complete exact check is the following pseudocode, where `isqrt` denotes the integer square root:

```
from math import isqrt

def J(x):
    return isqrt(x) if x % 2 == 0 else isqrt(x**3)

bad = []
for w in ("OOEOE", "OOOEE"):
    for n in range(2, 256):
        x = n
        realized = True
        for b in w:
            if ("E" if x % 2 == 0 else "O") != b:
                realized = False
                break
            x = J(x)
        if realized and x == n:
            bad.append((w, n))
print(bad) # []
```

Now suppose *OOOEE* is a cycle word at $n \geq 256$, and write $z = J^4(n)$ for the image after the prefix *OOOO*. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on *OOOO* gives $n^{81} \leq 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose *OOEOE* is a cycle word at $n \geq 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on *OOO* gives $n^{27} \leq 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters *OE* give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. We claim $(y+1)^3 < 2A^4$. If $y \leq A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \geq 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. $\square$

Theorem 3.6 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$. It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE . If $m \geq 5$ realizes OO , Lemma 3.4(i) gives $J^2(m) \geq (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E . Lemma 3.4(ii) and (iv) give $J^3(m) \geq (m+1)^2$ and $J^3(m) < (m+1)^2$.

Length five: the filter requires four odd letters among the first four, leaving only O^4E . Lemma 3.4(ii) and (iii) give $J^4(m) \geq (m+1)^2$, again contradicting (iv).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E , $EOOOOE$, $OEEOOE$, $OOEOOE$, and $OOOOEE$. For O^5E , Lemma 3.4(ii) and (iii) give $J^5(m) \geq (m+1)^2$, contradicting (iv). The word $EOOOOE$ rotates one step onto $OOOOEE$, and $OEEOOE$ rotates two steps onto $OOEOOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this word be a cycle word at some start, and rotate to a cycle minimum $m \geq 2$. The three alignments of the two even letters are $OOEOOE$, $OEEOOE$, and $EOEOOE$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE : the first image is even and strictly below m^2 , whereas every even state after a cycle minimum is at least m^2 , as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \geq 3$. The prefix OOE must be realized. If $m = 3$, then $3 \rightarrow 5 \rightarrow 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE . Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y , and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. $\square$

No exclusion of cycles of length seven or more is claimed. The census stops at length six because length seven admits even-terminating expanding words with two internal even letters that none of the recorded thresholds reach; no exclusion at length seven is claimed.

4. Short descent certificates

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$. The predicate is existential over all finite words, not only over words of length one or two.

Theorem 4.1 (uniform short certificates). Let $n \geq 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor \sqrt{n} \rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < n$ for $n \geq 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor \leq n^{3/4} < n$$

for $n \geq 2$. $\square$

The starts not covered by Theorem 4.1 are exactly the odd-to-odd starts: n odd and $J(n)$ odd. In particular, if $n \geq 2$ has no descent certificate of any length, then n is odd-to-odd. The converse is false: many odd-to-odd starts descend after a longer word.

If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1. The hypothesis is not proved. A certificate at n only reduces the problem to a strictly smaller positive integer, which may itself be odd-to-odd.

5. The remaining gap

Theorem 4.1 covers every even start and every odd-to-even start with a uniform one- or two-step certificate. The first odd-to-odd image expands, so ordinary strong induction cannot fire on the complement.

The gap resists a uniform run bound on expanding blocks. Lean certifies a chain of four consecutive expanding blocks,

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471.$$

Thus four consecutive expanding blocks between odd states occur. A uniform slack tax per expanding block is likewise impossible: the relative slack of a single letter tends to 0 with the state (Section 2), and the *OOE* example recorded there near $1.80 \cdot 10^{32}$ has relative slack below 10^{-30} .

No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether every start reaches 1, and open whether a nontrivial cycle of length seven or more exists.

6. Software archive

Lean proofs of the theorems of Sections 2–4 are in the project repository. The archive is not required to read the arguments above. From a clone, the review object for this note is

```
cd formal && lake build Problems.JugglerPaper
```

That barrel imports only the modules named by this note (Appendix A).

Appendix A. Lean names

The proofs in the text are ordinary integer arguments. The following names are the corresponding Lean theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`.

Text	Lean
itinerary semantics	<code>follows_iff_word</code> , <code>image_eq_iterate</code> , <code>image_append</code>
Theorem 2.1	<code>image_monotone_of_follows</code>
Theorem 2.2	<code>power_bound_word</code>
Corollary 2.3	<code>power_bound_contracts</code>
Theorem 2.4	<code>global_defect_identity</code>
Theorem 2.5	<code>global_defect_eq_zero_iff_localsTight</code> , <code>global_defect_eq_zero_implies_monochrome</code> , <code>power_bound_eq_iff_extremal</code>
Theorem 2.6	<code>global_defect_append</code>
Corollary 2.7	<code>image_eq_start_defectRatio</code>
per-step slack	<code>one_plus_eta_lt_succ_sq</code>
Lemma 3.1	<code>odd_cell_unique</code>

Text	Lean
Theorem 3.2	<code>cycle_word_formally_expanding,</code> <code>cycleMin_not_end_odd,</code> <code>square_scale_superquadratic,</code> <code>cycleMin_to_even_superquadratic</code>
Lemma 3.3	<code>lower_growth_word</code>
Lemma 3.4	<code>oo_suffix_threshold,</code> <code>ooo_suffix_threshold,</code> <code>threshold_inherits_odd_append</code>
Lemma 3.5	<code>no_cycle_word_oooeoe,</code> <code>no_cycle_word_ooooee</code>
Theorem 3.6	<code>no_cycle_word_length_le_six</code>
Theorem 4.1	<code>even_finiteProgress,</code> <code>odd_even_finiteProgress</code>
no certificate $\Rightarrow$ odd-to-odd	<code>no_finiteProgress_implies_odd_odd</code>
induction to 1	<code>reachesOne_of_all_finiteProgress</code>
four-block chain	<code>four_block_pe_1999</code>

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I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. Lean theorems and named computations are the certificates for the claims of Sections 2–4. I take full responsibility for the contents.

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